TESR Co., Ltd.
THAI EMBEDDED SYSTEM AND ROBOTICS

TESR ROBOTICS · USER MANUAL V1.0

Speedybot Delivery Robot — User Manual

Setup · Mapping · Map editing · Adding points · Running tasks · Testing — step by step with video tutorials



Read online (with videos)
https://tesr-channel.github.io/Speedybot_API/manual.html
All videos folder:
Google Drive

Updated 2026-10-15 · v1.0

Contents

00 **Before you start**

01 **Hardware setup & power**
- 01 · Basic setup

02 **Mapping**
- 02 · Mapping

03 **Edit the map & set it as default**
- 03 · Edit map

04 **Navigation — add points**
- 04 · Navigation — add points

05 **Navigation — robot run**
- 05 · Navigation — robot run

06 **Testing & hand-over**
- 06 · Outdoor field test

1 Setup | 2 Mapping | 3 Edit map | 4 Add points | 5 Run tasks | 6 Test

00

Before you start

Speedybot is an indoor/outdoor delivery robot with lockable compartments, LiDAR navigation and automatic obstacle avoidance. This manual walks through everything from unboxing to running real deliveries in six sections, in the order you do them. Each section has a video tutorial.



Product specifications



Core features

What you need

- Hex (Allen) key to remove the side cover and the board
- An activated 4G/5G data SIM card
- The robot battery and charger (manufacturer parts only)
- A WiFi tablet or computer for mapping (the vendor notes Apple devices are not supported yet)
- A smartphone for the H5 web app
- The admin account issued by TESR — used for both the Deployment page and the H5 app

Your robot's access details

Robot SN	
Robot WiFi hotspot	`SpeedyBot + SN`
WiFi password (factory)	`dnk131212`
Admin username	
Password	
Mapping page (on robot WiFi)	`http://192.168.99.2:3578/alpha-pro-scan/index.html`
H5 operation app	`https://alpha-pro.csjbot.com`

Fill in and keep safe · do not share the account · contact TESR for resets

⚠ Safety first

- Always remove the battery before removing or refitting the circuit board
- Do not push the robot while it is powered on and working — release the hub first, and push no faster than 30 m/min
- Never block the LiDAR on top of the robot during mapping or tasks
- Use remote control only when the robot is in sight, nobody is nearby and the path is clear
- Do not load hazardous goods, spill-prone liquids or items beyond the compartment size/weight
- Do not disassemble the robot or alter wiring; contact TESR for repairs

01 Hardware setup & power

Insert the SIM, fit the battery, close the side cover, power on/off and charge

▶ VIDEO TUTORIAL

01 · Basic setup

https://drive.google.com/file/d/1Kf488uSSo0MfIbchaxYraZx1Jy9lO2II/view

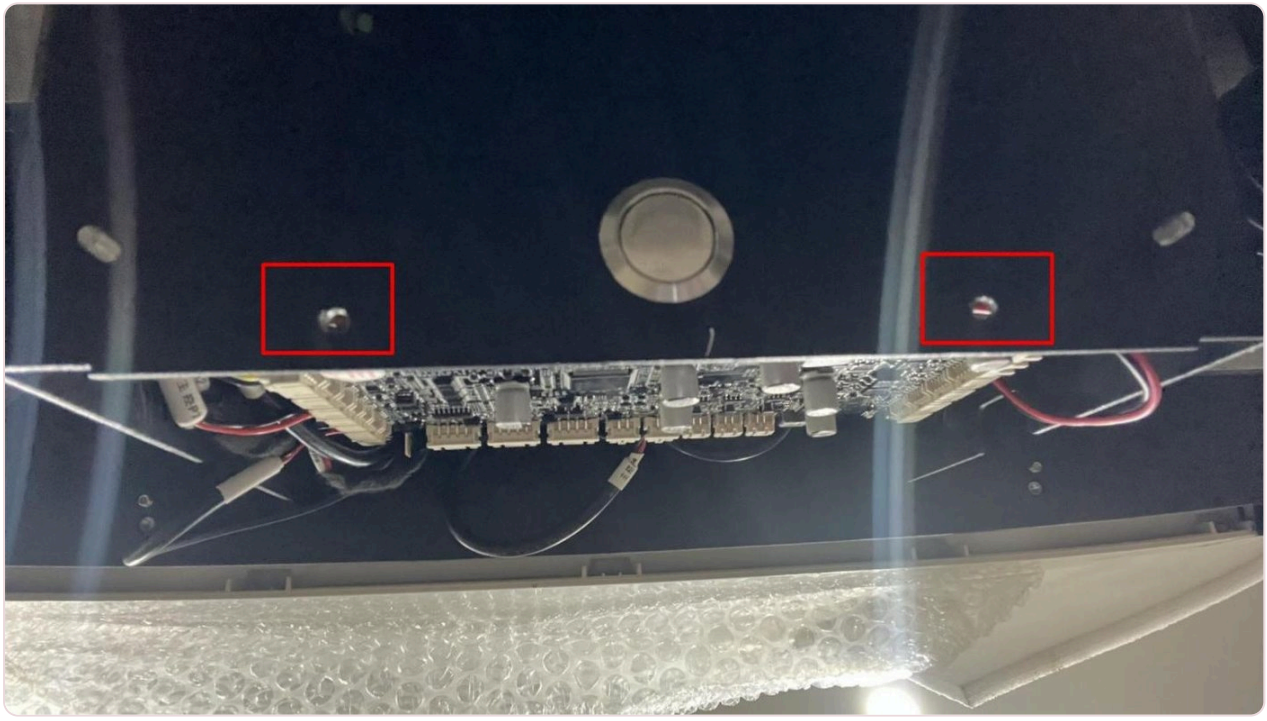
Scan to watch

1.1 Insert the SIM card

1

Remove the side cover

Use the hex key to remove 4 screws (2 on each side), then lift off the side cover.



Side cover screws

2

Remove the battery first

The battery must be removed **every time** before you remove the circuit board.

> ⚠ **Caution** Never remove or fit the board with the battery connected

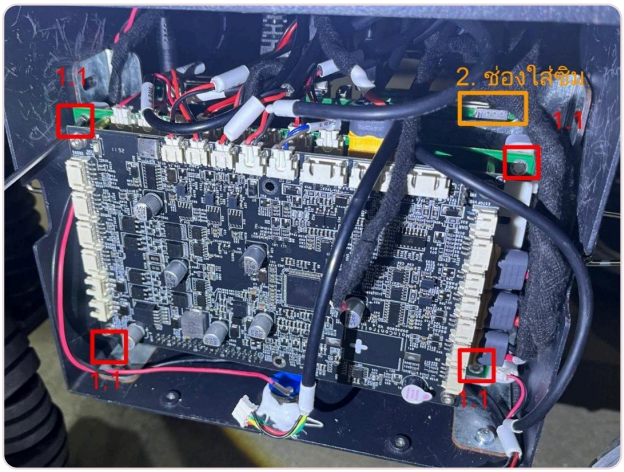


Removing the battery

3

Remove the green board

Unscrew the 4 screws holding the green board and lift it out so the SIM slot is easy to reach.

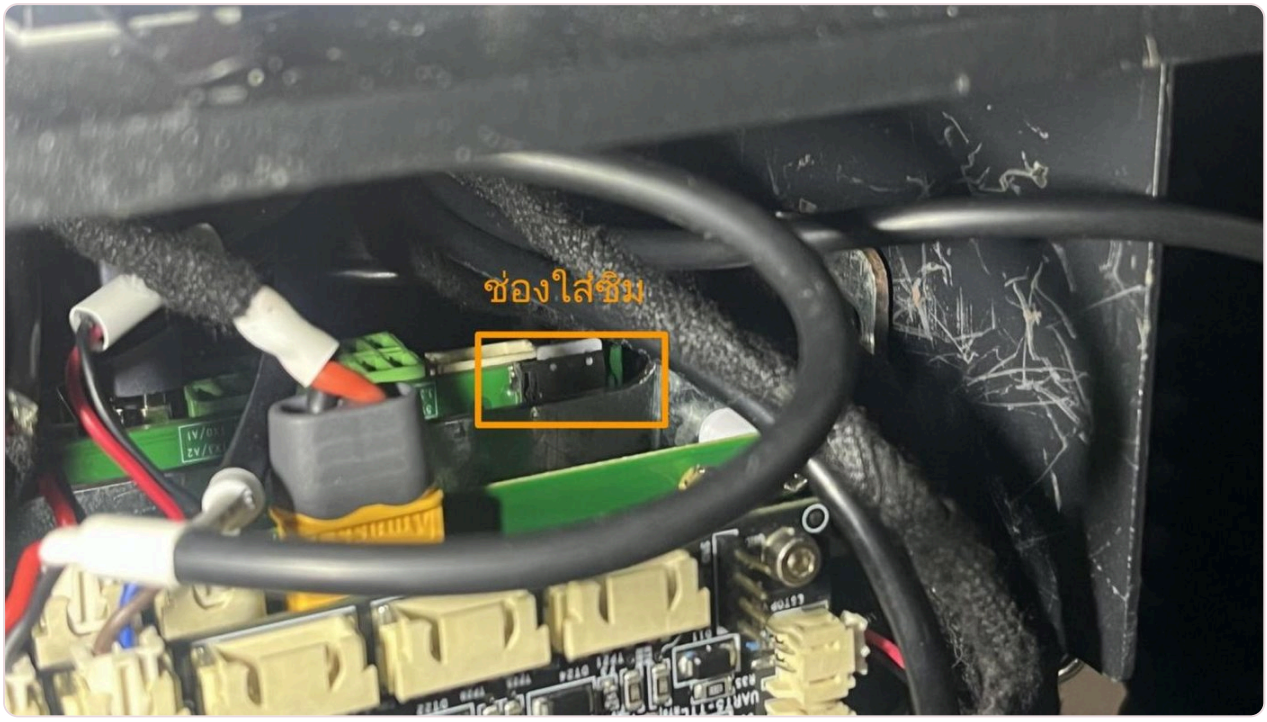


The 4 board screws and the SIM slot

4

Insert the SIM and refit

Push the SIM fully into the slot, then screw the green board back in place.



SIM slot close-up

1.2 Fit the battery and close the cover

1

Insert the battery

Slide the battery fully into its bay and lock it (reverse the steps to remove it).



Inserting the battery

2

Refit the side cover

Align the cover and tighten all 4 screws.



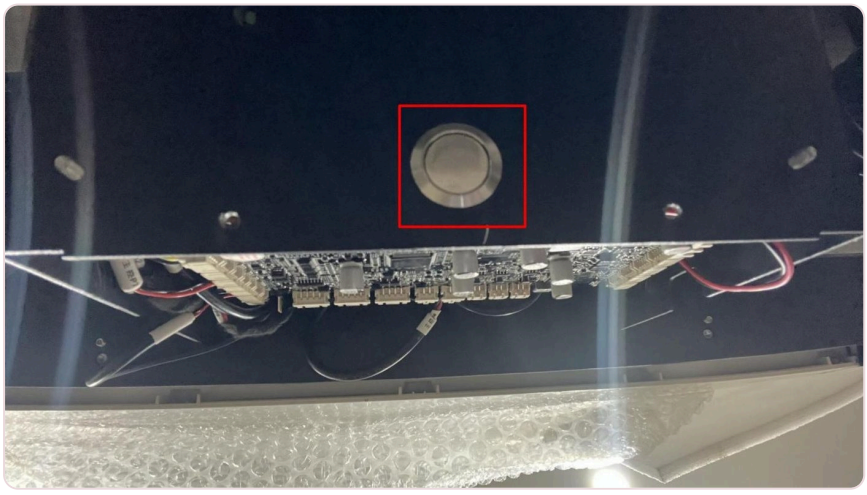
Side cover refitted

1.3 Power on / off

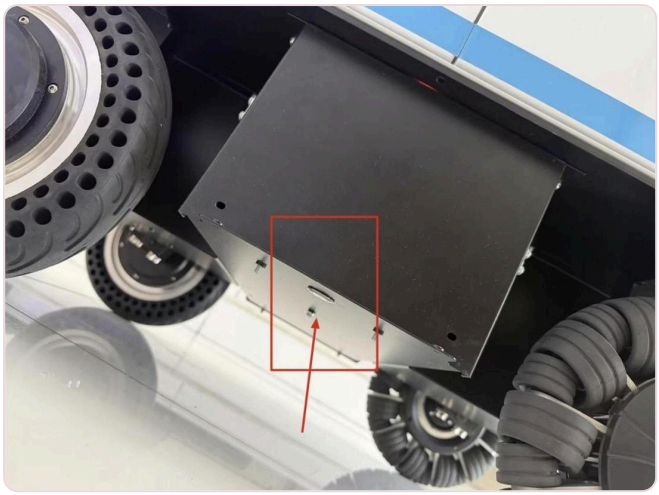
1

Power on

Press the silver power button under the robot for about 1 second until it beeps once. Booting takes about 5 minutes.



Power button under the robot



Switch location (vendor manual)

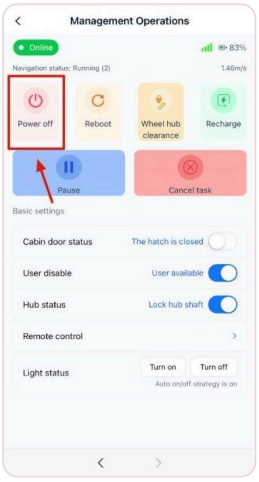
2

Power off

Method 1: press and hold the power button for 10 seconds.

Method 2: in the H5 app open **Management Operations** → **Power off** and confirm.

> **Tip** Only power off when the robot is idle and the area is safe



Power off from the H5 app

1.4 Charging

> **Note** The current software does not yet support automatic return-to-dock. Push the robot to the charger manually.

1

Release the hub and push to the charger

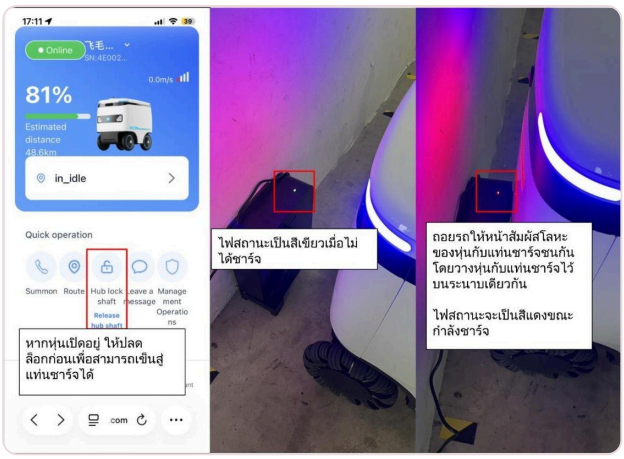
In the H5 app tap **Release hub shaft** so the robot can be pushed. After charging tap **Lock hub shaft** again before starting tasks.

2

Plug in the charger

Plug the charger into mains first, wait 2–3 seconds, then connect it to the charging port at the lower rear of the robot.

Charger light: **red** = charging · **green** = full / idle.



Releasing the hub and charger lights

> ⚠ **Caution** Use only the manufacturer's battery and charger · when stored, fully charge at least once a month · if it will not charge or the light is abnormal, stop and contact TESR

02

Mapping

Join the robot's WiFi, open the Open Deployment Platform, create a map and walk the robot through the whole area

▶ VIDEO TUTORIAL

02 · Mapping

https://drive.google.com/file/d/13CF4OGyT-PtTbzVsXLsD0kw0piT98234/view

Scan to watch

Important Key rule: before you start, fix the **origin and the robot's heading** · walk with the robot the whole time and keep WiFi stable · finish back at the origin with the same heading

2.1 Prepare the origin

1 **Choose the start point**

With the robot off, push it to the start point. Pick a quiet spot that is easy to return to (avoid the middle of a lobby).

2 **Photograph the position and heading**

Take a photo of where the robot stands and which way it faces so you can put it back exactly. This is where the robot **localizes every time it powers on**.

3 **Power on and wait**

Power on at this spot and wait about 5 minutes for the system to be ready.

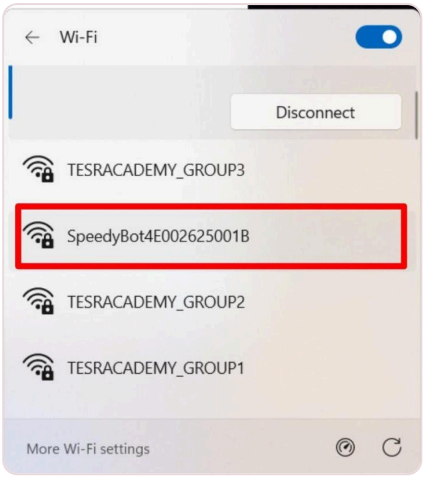
2.2 Join the robot WiFi and log in

1

Join the robot's hotspot

On the tablet or computer, join `SpeedyBot` + serial number (e.g. `SpeedyBot4E002625001B`), password `dnk131212`.

> ⚠ **Caution** Stay with the robot and keep at least 2 bars of WiFi. A long disconnection may force you to remap.

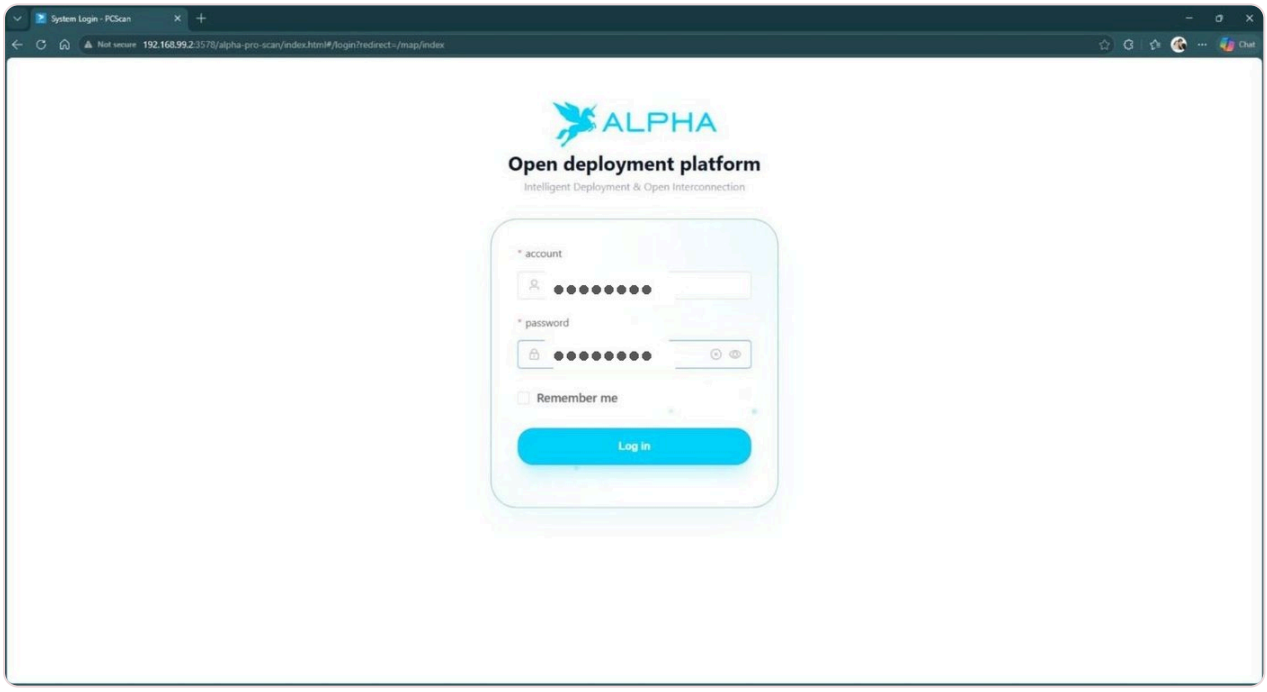


Selecting the robot's WiFi

2

Open the Open Deployment Platform

In a browser open `http://192.168.99.2:3578/alpha-pro-scan/index.html` and log in with the admin account from TESR.



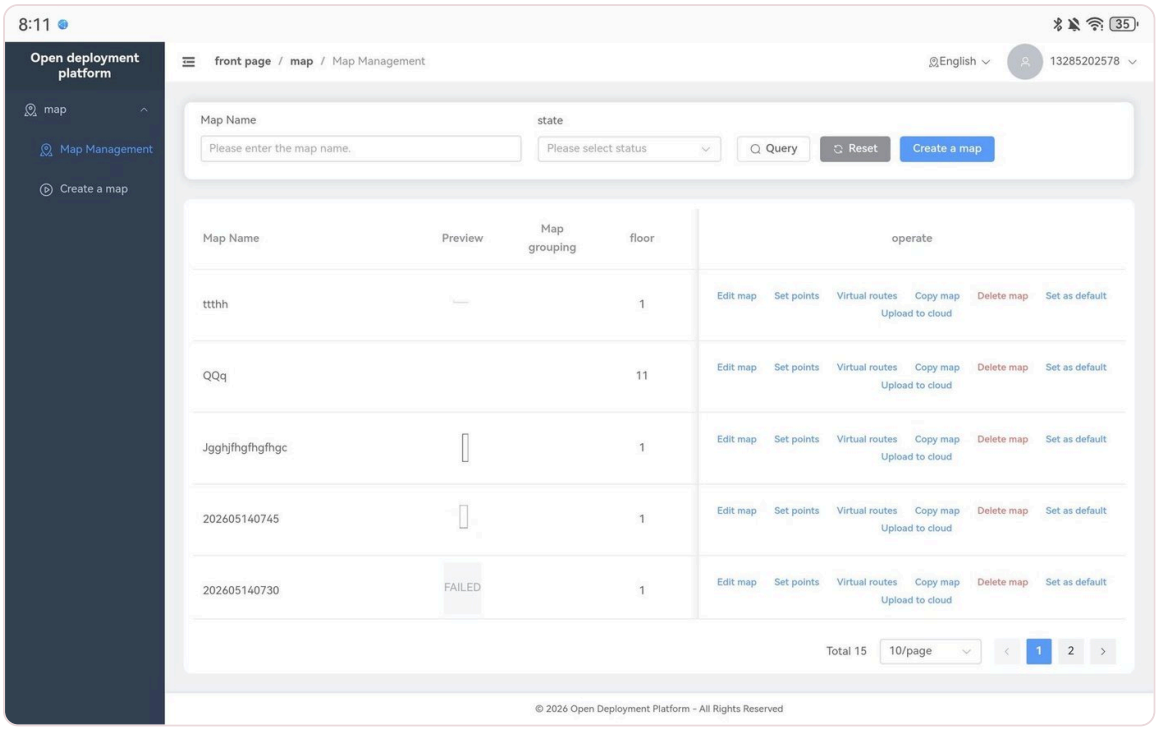
Login page (account hidden)

2.3 Create a new map

1

Click Create map

Go to **Map management** and click **Create map** at the top right.



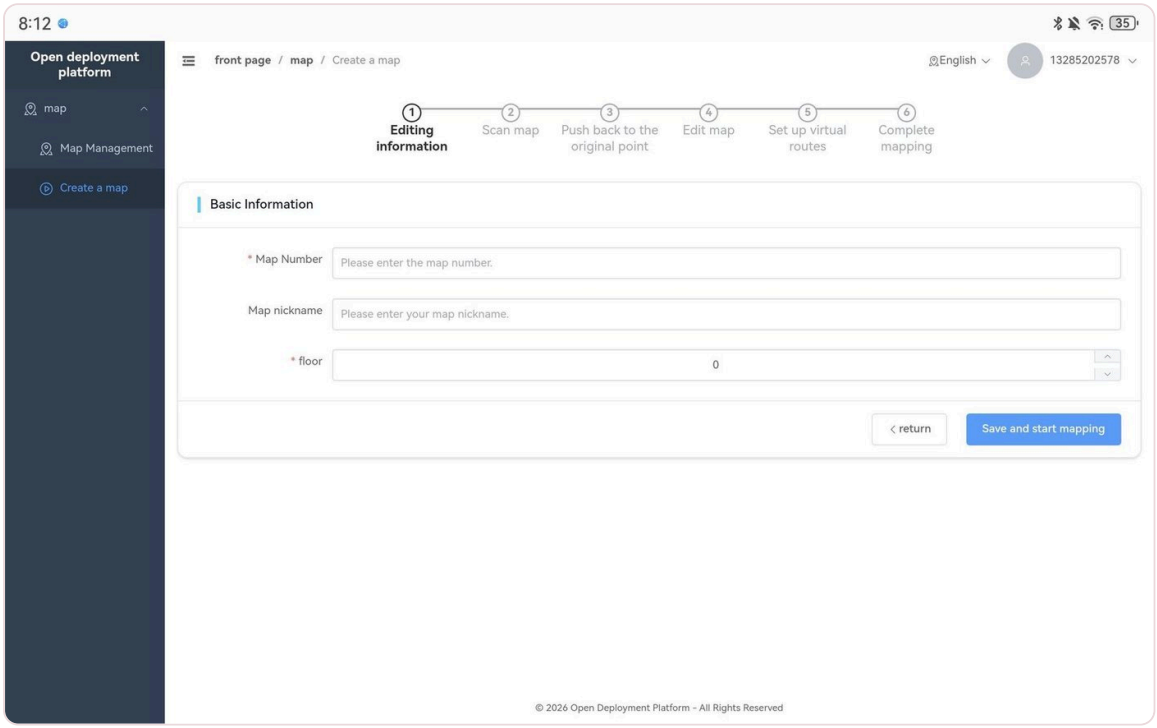
Map management and the Create map button

2

Fill in the map details

Fill in the map number, name and floor as in the table below, then click **Save and start mapping**.

Field	Rule	Example
Map number (ID)	Letters, digits and _ only; no spaces or Thai/Chinese; max 10 characters; unique	A_ZONE_4F
Map nickname	Any language; easy to recognise on site	Building A 4F
Floor	A number; used later for elevator integration	4

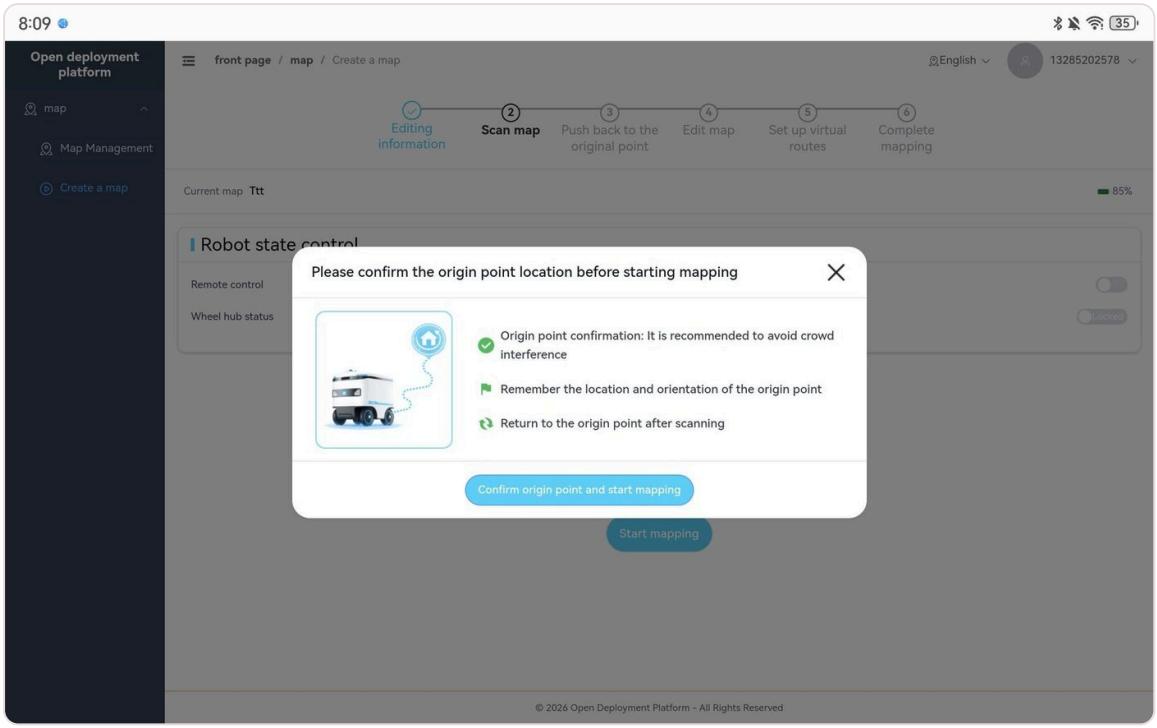


Map information form

2.4 Scan the map

1 Confirm the origin

Check the robot is on the origin with the recorded heading, then click **Confirm origin and start mapping** (the wheels stay locked until the system is ready).

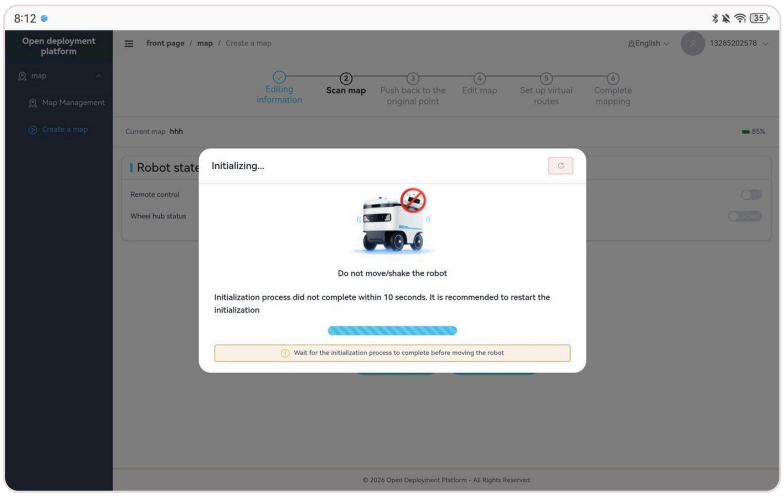


Confirm-origin dialog

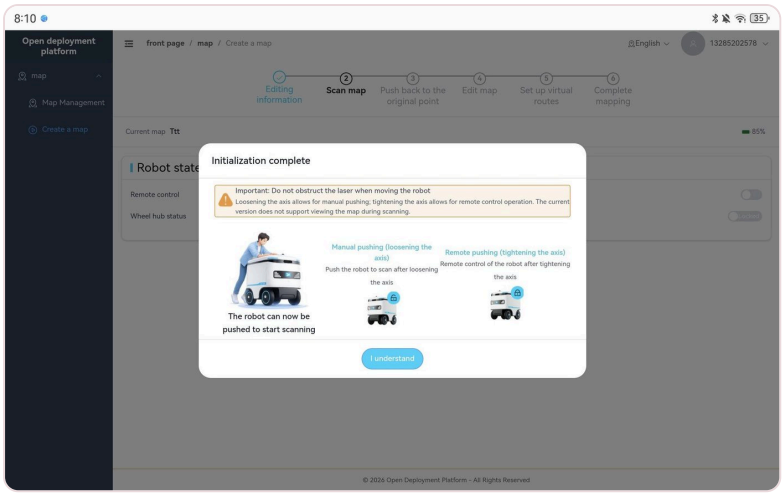
2 Wait for initialization

Initialization takes up to ~10 seconds. **Do not push or drive the robot** until the ready message appears.

> **Tip** If nothing happens after 10 s and other buttons are inactive, restart the robot at the origin, wait ~5 min and start again



Initializing



Initialization complete — you can move the robot

3

Cover the whole area

Indoors: push at a steady pace through every corridor, room and turn the robot will use.

Outdoors: you may use remote control (lock the hub first; keep defaults angular 0.50 / linear 0.8) and stay within 20 m of the robot.

> ⚠ **Caution** Do not block the LiDAR · this version cannot show the live map while scanning, so cover every route on foot

2.5 Finish mapping

1

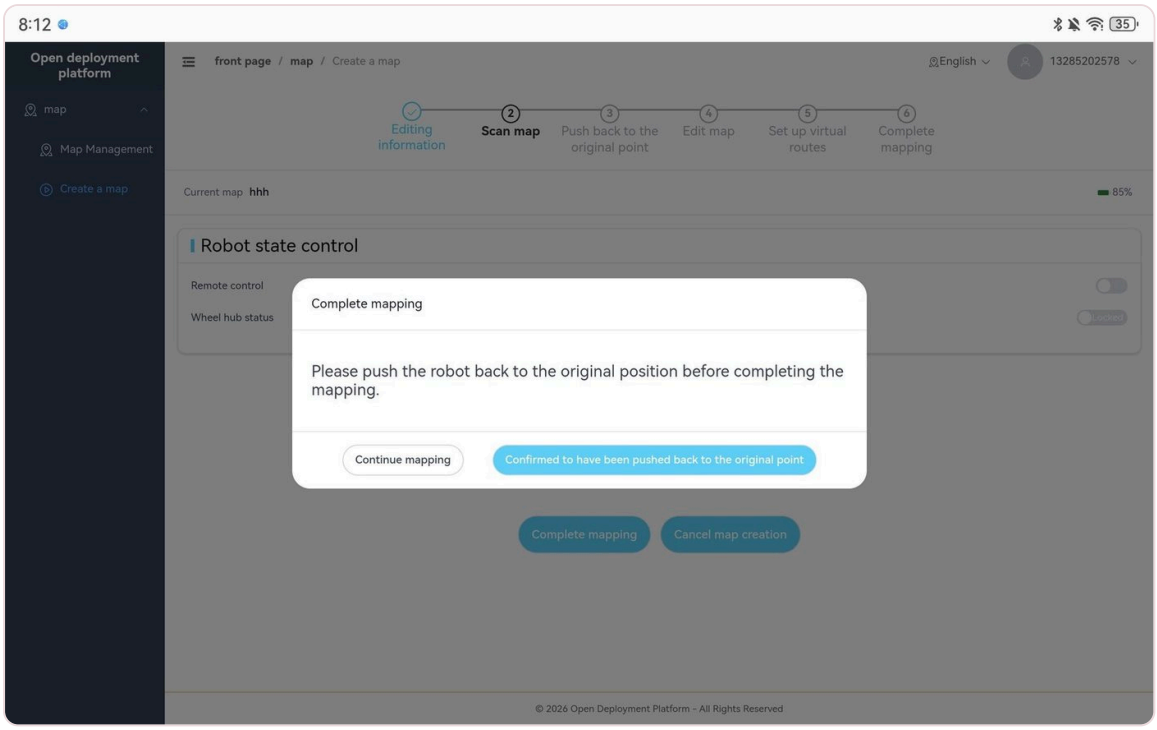
Return to the origin

Push the robot back to the origin within 20 cm and with the same heading as at the start.

2

Confirm return to origin

Click **Confirm returned to origin** and wait for processing. Do not close the page or drop WiFi. The map editor opens automatically when done.



Finish-mapping dialog

03 Edit the map & set it as default

Erase noise, draw virtual walls/routes, save, then sync and set the default map in the H5 app

▶ VIDEO TUTORIAL

03 · Edit map

https://drive.google.com/file/d/1giYCBIMVQR7IzrefL58bDnhaG8VScCPE/view

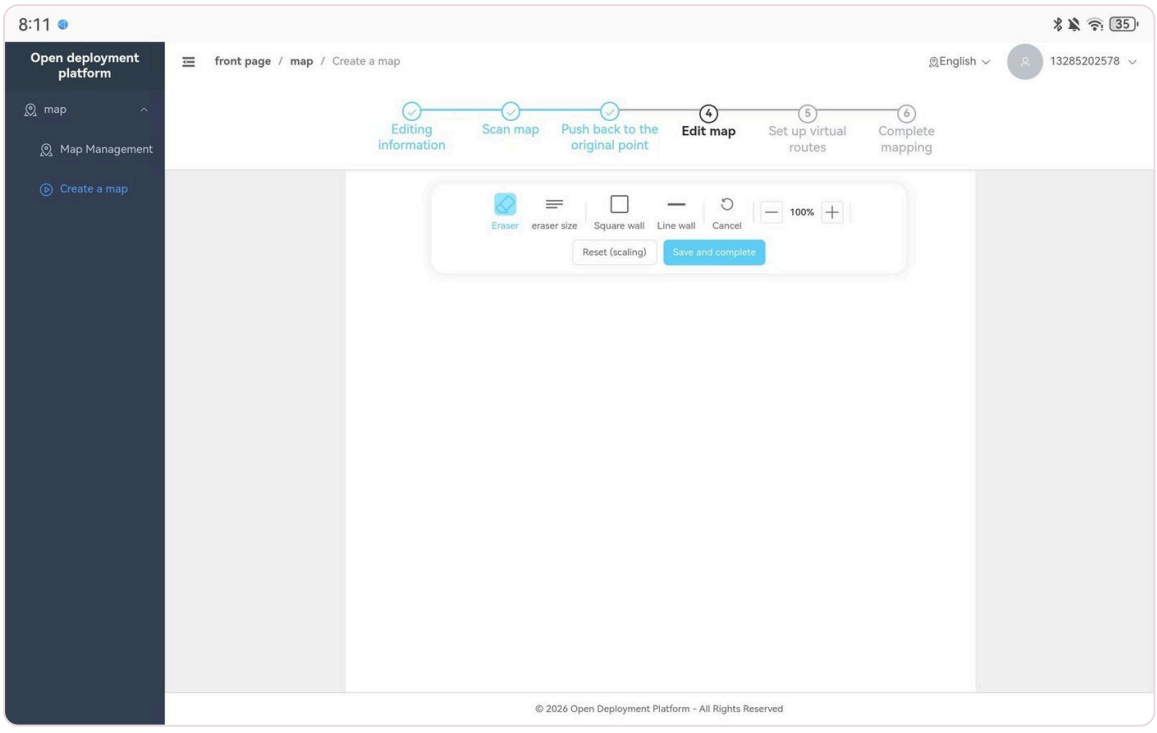
Scan to watch

3.1 Edit on the Deployment page (tablet recommended)

1

Open the editor

It opens automatically after mapping; otherwise click **Edit map** on the map in the list.
Two-finger drag to pan · pinch to zoom · **Reset 100%** to restore zoom.



Map editor toolbar

2

Remove noise with the Eraser

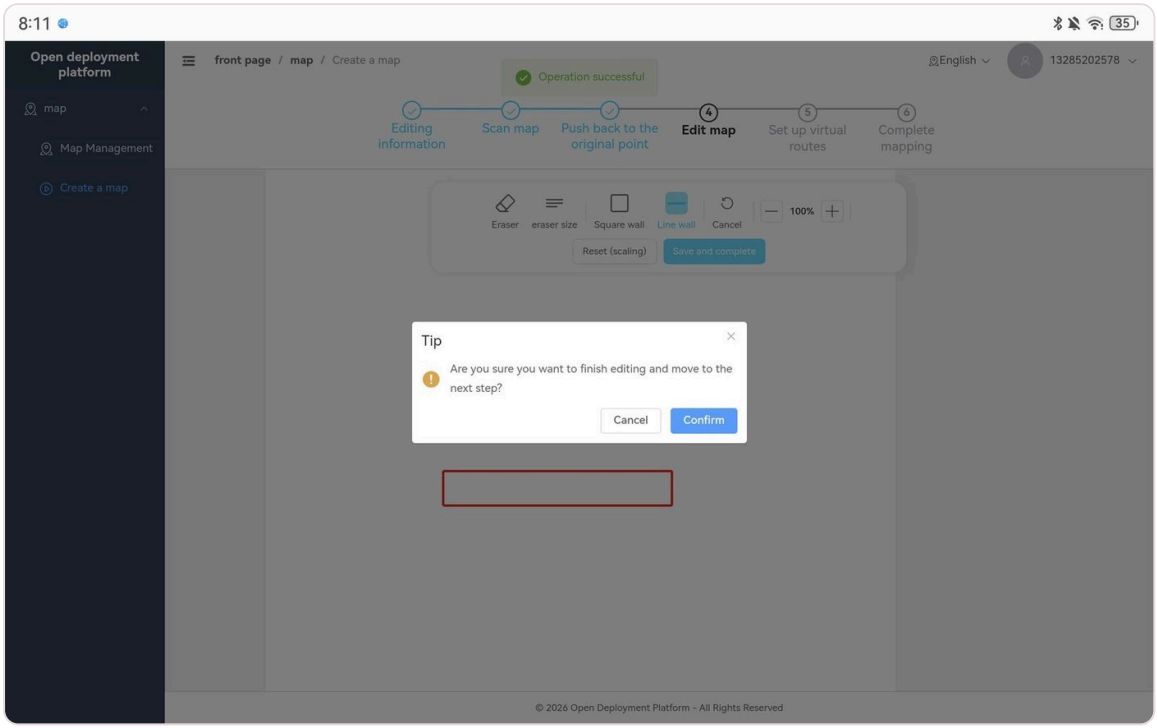
Choose **Eraser** → pick a size (small/medium/large) → swipe with one finger over the noise.

3 Draw virtual walls

Line wall: tap points to draw a polyline, then Finish (for separate segments, click Finish drawing after each).

Rectangle: drag a box, then Finish.

The robot never crosses a virtual wall — use it for stairs, kerbs and no-go areas.



Confirming a virtual wall

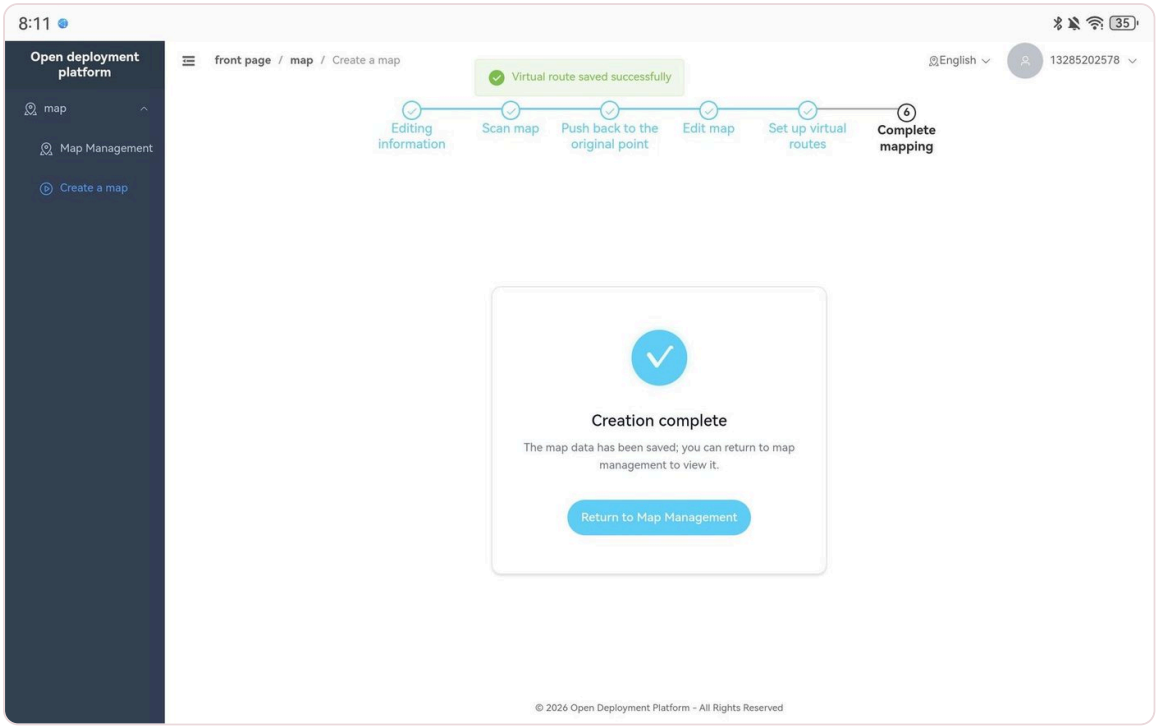
4 Draw virtual routes (optional)

Zoom in and brush the routes the robot should follow. Multiple routes must connect (at least one intersection). Skip if not needed.

5

Save

Click **Save and finish editing** and wait for the success message.



Map creation complete

3.2 Log in to the H5 app

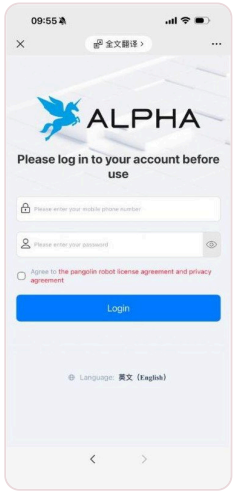
1

Open the H5 app

Scan the QR code on the robot or open `https://alpha-pro.csjbot.com`, log in with the admin account (same as the Deployment page), accept the agreement and tap Login.



QR code on the robot



H5 login page

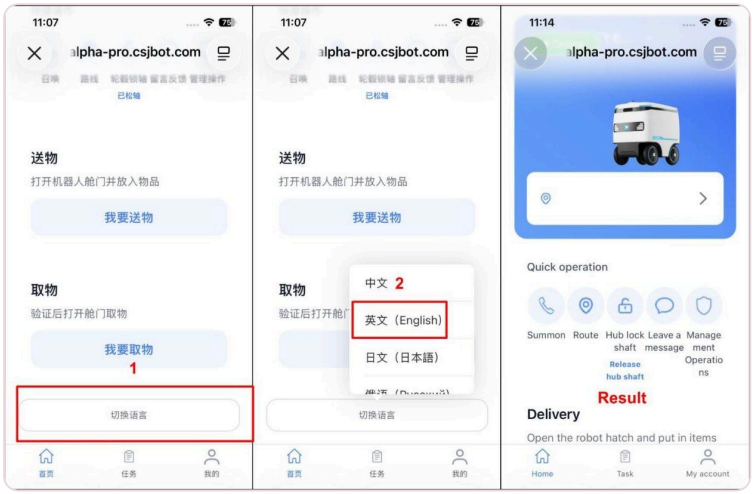
2

Switch the language to English

Tap **Language** at the bottom of the login page and choose English; after login choose it again at the bottom of Home.



Language before login



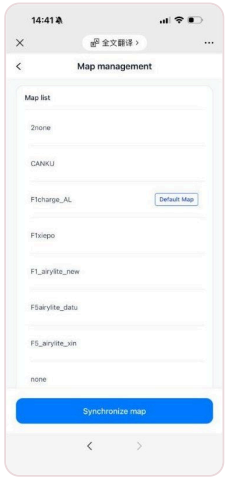
Language after login

3.3 Sync the map and set it as default

1

Synchronize maps

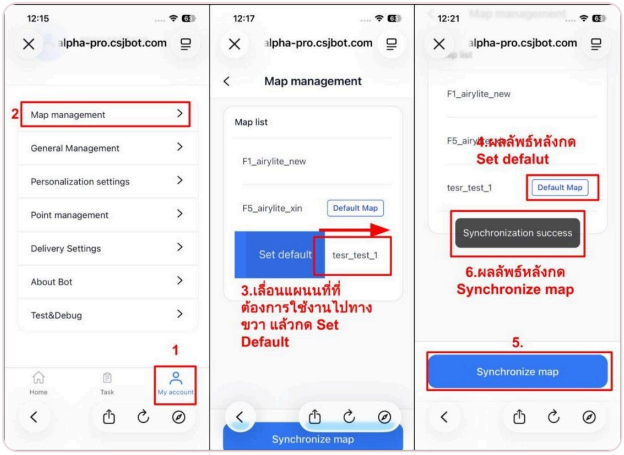
Go to **My account** → **Map management** and tap **Synchronize map**; wait for “Synchronization successful”.



Map list and Synchronize map

2 Set as default

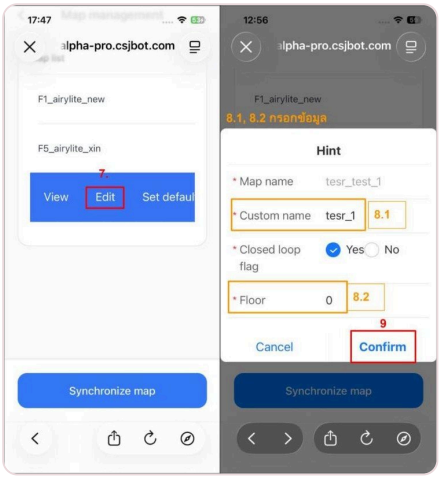
Pick the map, swipe right and tap **Set default**. It now shows a **Default Map** badge.



Steps 1–6: sync and set default

3 Rename / set floor (optional)

Edit the map to set its display name and floor, then save.



Steps 7–9: name and floor

4 Return to origin and restart

Release the hub, push the robot back to the **map origin** (same heading) and **power-cycle** it. The robot loads the default map and localizes from there.

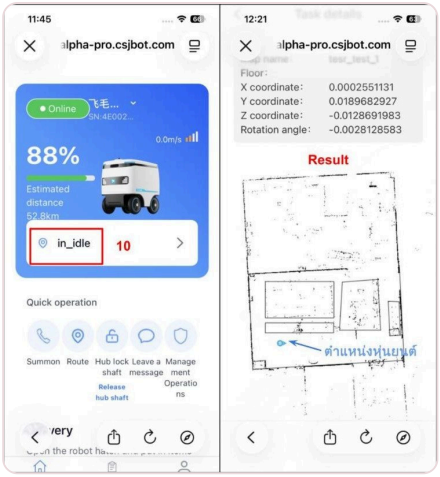
> ⚠ **Caution** Setting a default map does not switch maps immediately — always restart the robot at that map's origin

5

Check the map

When the robot is ready (**in_idle**), tap the status card on Home to see the map and robot position; check it matches reality.

> **Tip** You can also watch the live map and position on the TESR Speedybot API Lab dashboard



Checking the map from the status card

04 Navigation — add points

Record delivery, waiting and special points, then sync them so the robot can route between them

▶ VIDEO TUTORIAL

04 · Navigation — add points

https://drive.google.com/file/d/1xfZTQYDCSzyhE4_aFloX-QBX4snaxaxr/view

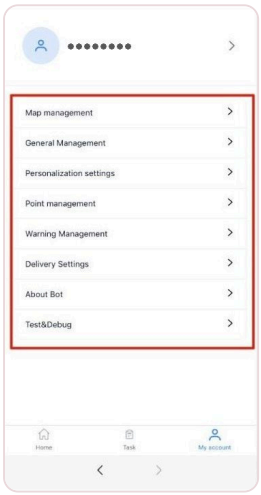
Scan to watch

4.1 Common points

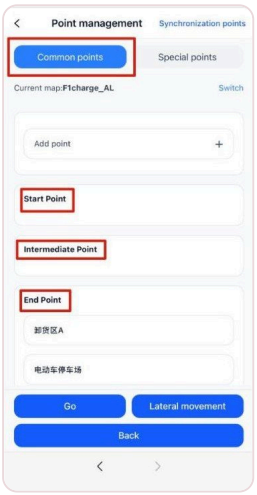
1

Open Point management

Go to **My account** → **Point management**, choose **Common points** and check **Current map** is the active map (tap Switch to change).



Admin menu in My account



Common points: Start / Intermediate / End

2

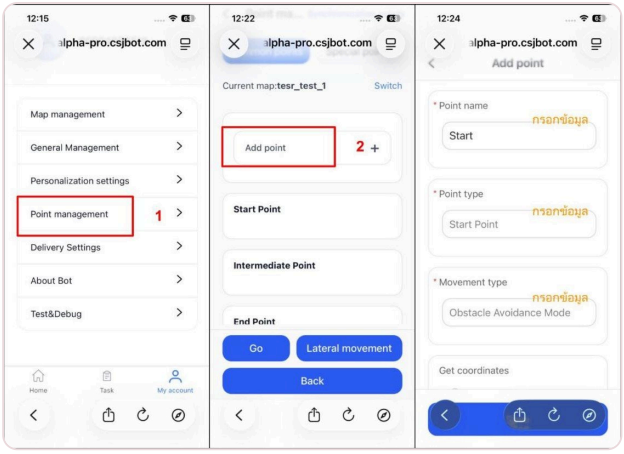
Push the robot to the spot

Release the hub and push the robot to the place you want to save (e.g. a door or pickup spot), facing the way it should park.

3

Tap Add point +

Enter the **point name** and **Point type**: Start, Intermediate or End point.

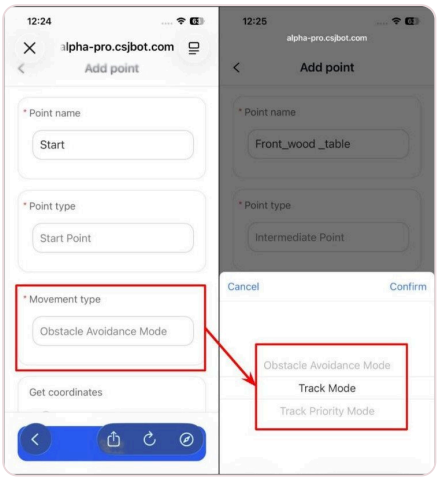


Adding a point

4

Choose the movement type

Options are Obstacle avoidance / Track / Track-priority — **use Obstacle Avoidance for now**.

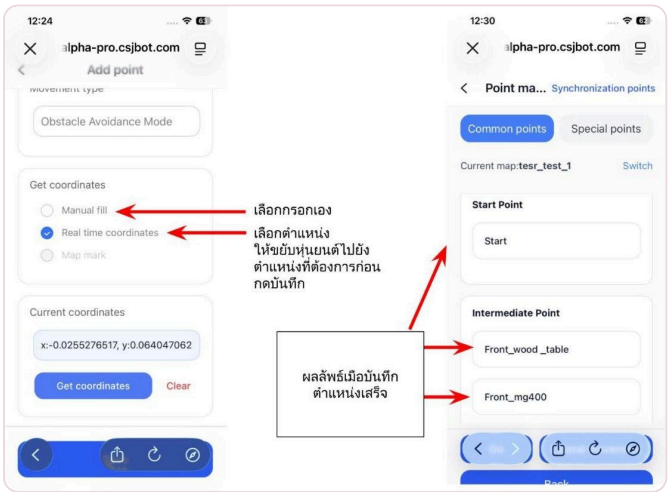


Movement type options

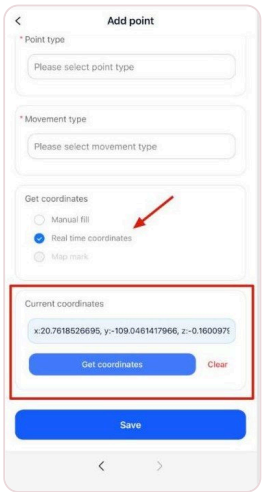
5

Set the coordinates

Real time coordinates (recommended): tap **Get coordinates** to use the robot's current position.
Manual fill: type X, Y, Z and rotation (e.g. origin = 0, 0, 0).



Manual vs real-time coordinates



Get coordinates

6

Save

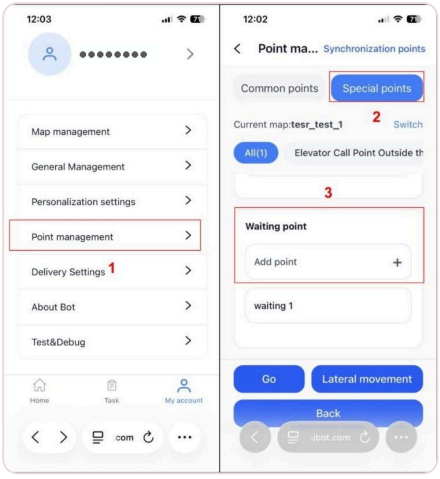
Tap **Save**. Repeat for every point.

4.2 Special points

1

Waiting point

Special points tab → **Waiting point** → **Add point**. This is where the robot goes after finishing or cancelling a task.

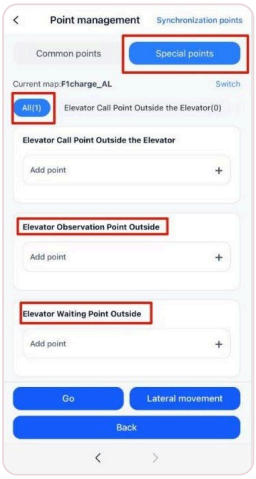


Setting a waiting point

2

Elevator points

Elevator call point / observation point / waiting point outside the elevator — for elevator integration, set per site.



Special point types

4.3 Sync and test the points

1

Tap Synchronization points (important)

After adding or editing points, always tap **Synchronization points** (top right) to upload point connectivity. Without it the robot cannot route between points.

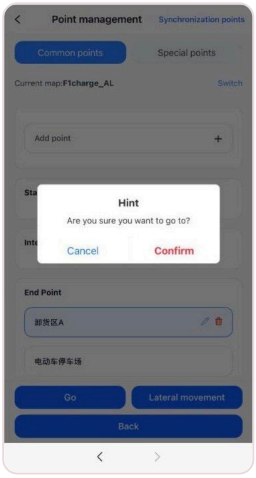


Synchronization points button

2

Lock the hub and test Go

Lock the hub, select a point and tap **Go** → **Confirm**; the robot drives there (Lateral movement / Back nudge the position).



Confirm Go

3

Edit or delete

Select a point and tap the pencil to edit or the bin to delete. Check no route or delivery uses it, then sync again.

05 Navigation — robot run

Pre-run settings, summon, deliver, pick up, track tasks and use the admin controls

▶ VIDEO TUTORIAL

05 · Navigation — robot run

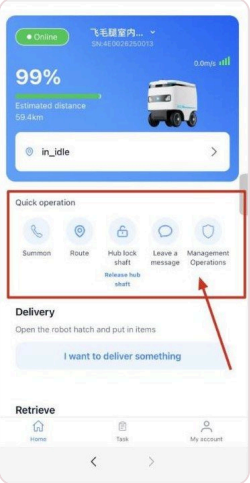
https://drive.google.com/file/d/1_w0PwMQ-si4WrHpa9ZKcgraM39YFk34C/view

Scan to watch

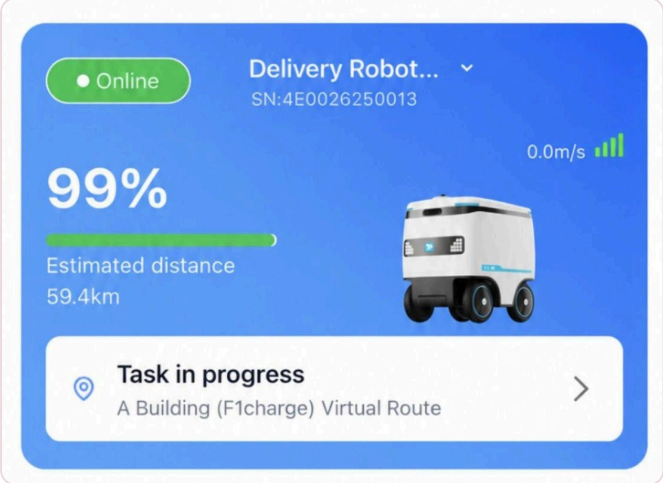
5.1 Home and robot status

1 Read the status card

The top card shows the robot name, SN, online state, battery %, estimated range, speed, signal and task status (in_idle = idle).



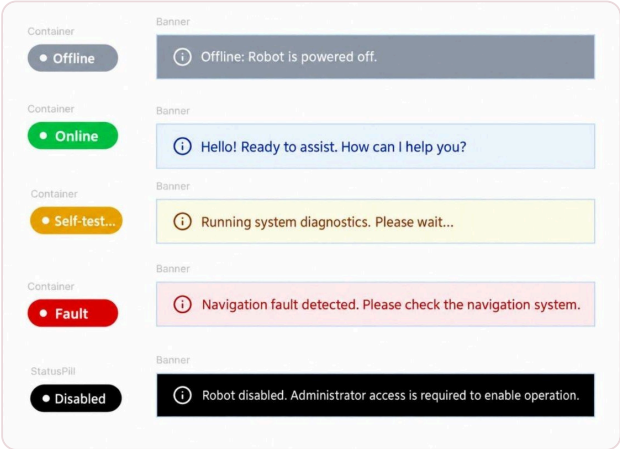
Admin home page



Status card during a task

2 Status meanings

Green Online ready · **Grey Offline** check power/network · **Yellow Self-testing** wait · **Red Fault** stop tasks and call an admin · **Black Disabled** regular users cannot operate.

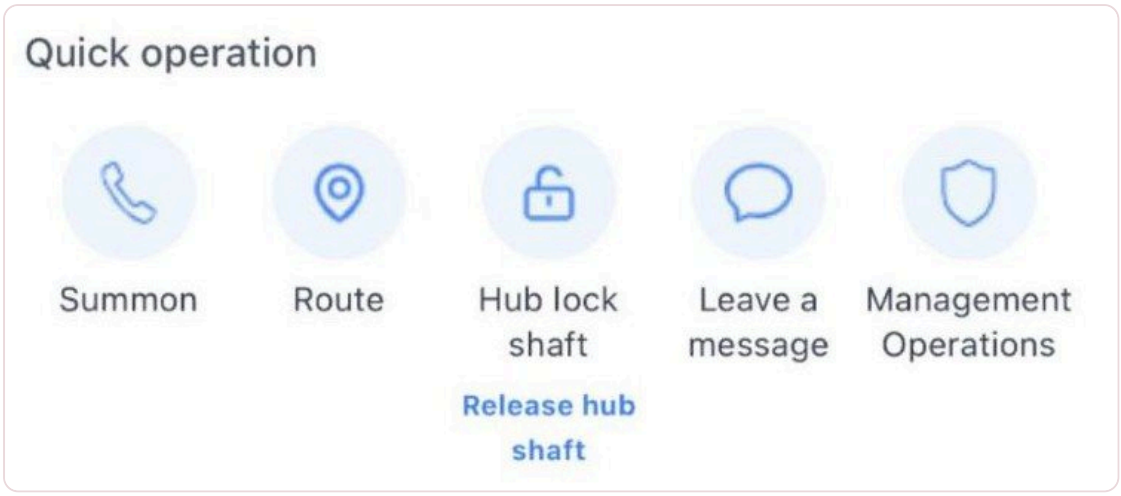


Status indicators

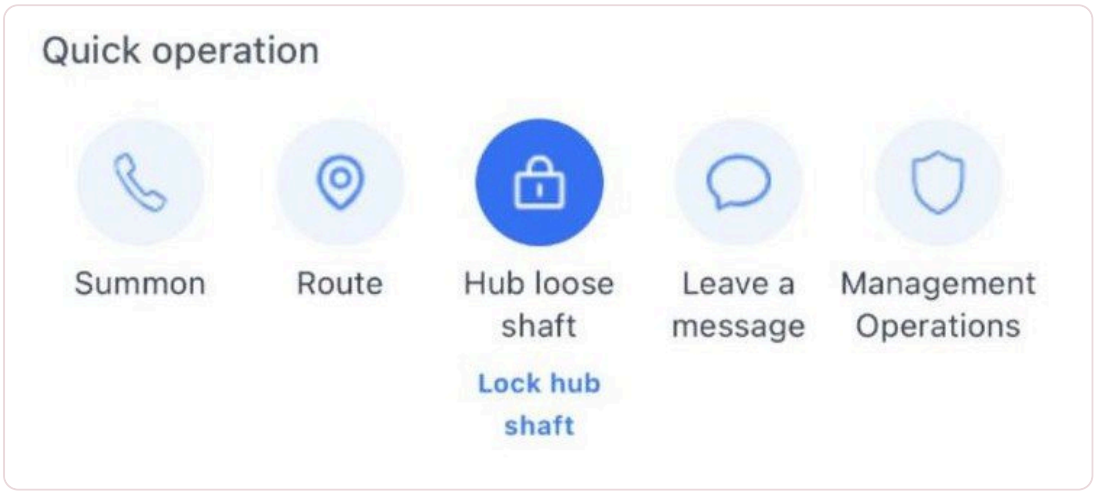
3

Lock / release the hub

In Quick operation: **Lock hub shaft** = ready for tasks · **Release hub shaft** = can be pushed by hand (do not start tasks while released).



Hub locked



Hub released

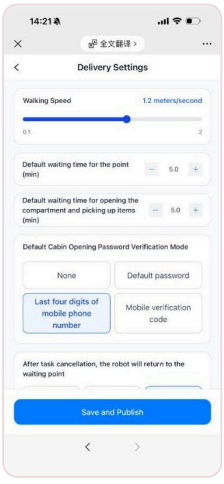
5.2 Pre-run settings

1

Speed and wait times (Delivery settings)

My account → **Delivery settings**: set walking speed (0.1–2 m/s), point wait time, compartment pickup wait time and the opening verification method, then **Save and Publish**.

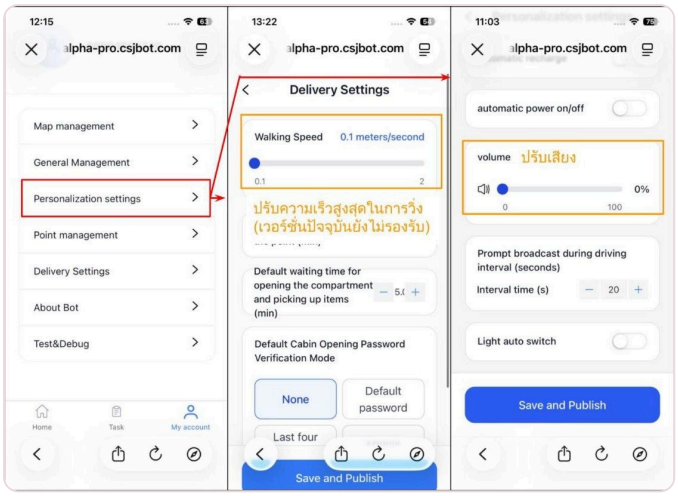
> **Tip** Start testing at low speed (e.g. 0.3–0.5 m/s), then increase



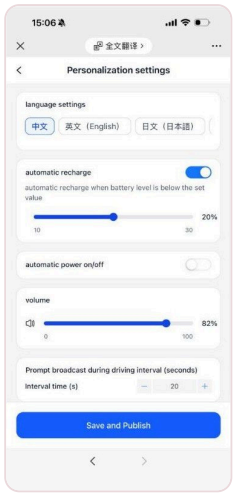
Delivery settings

2 Language, volume and power (Personalization)

My account → Personalization settings: language, automatic recharge (not supported in this version yet), scheduled power on/off, volume and voice-prompt interval, then **Save and Publish**.



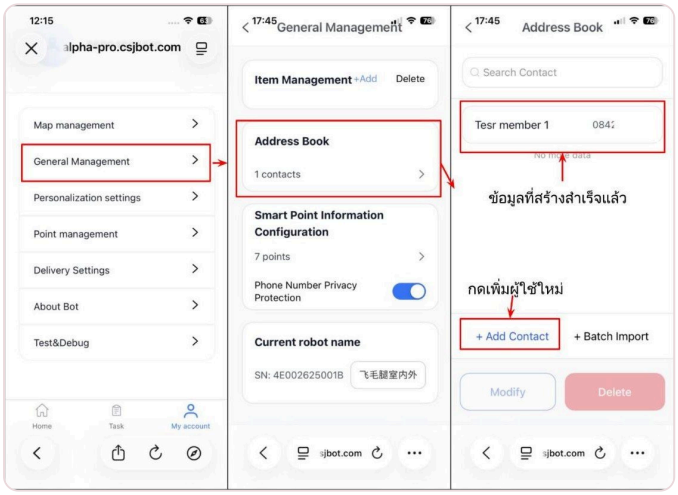
Volume (speed here is not supported — use Delivery settings)



Personalization settings

3 Address book

My account → General management → Address book → + Add contact: name and phone (Excel import available). Bind up to 3 recipients per point in Smart point information.



General management → Address book

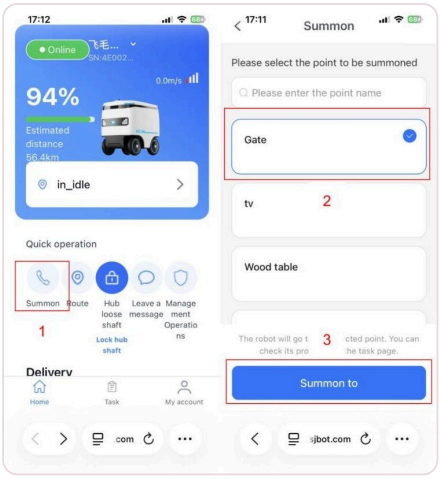


Add contact form

5.3 Summon

1 Pick the point

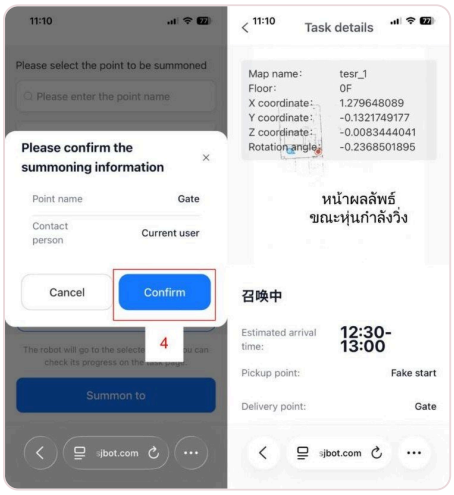
On Home tap **Summon** → select (or search) the point → check the notification contact.



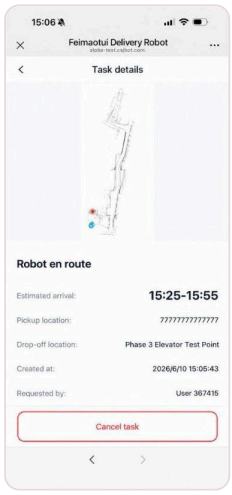
Summon steps 1–3

2 Confirm

Tap **Summon to** → **Confirm**; the robot drives to the point. Follow progress in the Task tab.



Confirmation and result



Task details en route

3 Cancel

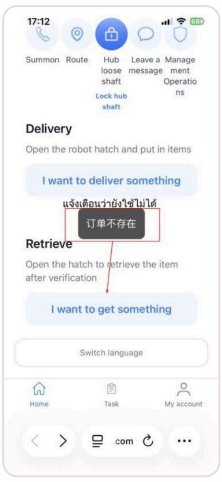
Tap **Cancel task** in the task details; the robot returns to the admin-set return point.

5.4 Send an item (Delivery)

> ⓘ **Note** You must **create a delivery first** ; tapping Retrieve before that shows an error.

1 Open Delivery

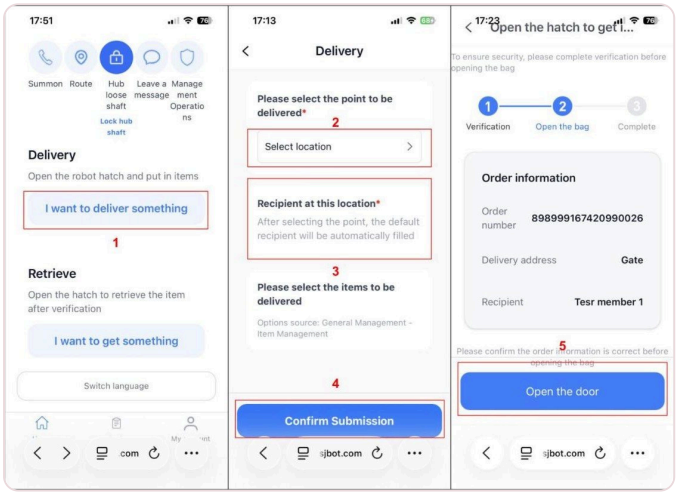
On Home tap **I want to deliver something**.



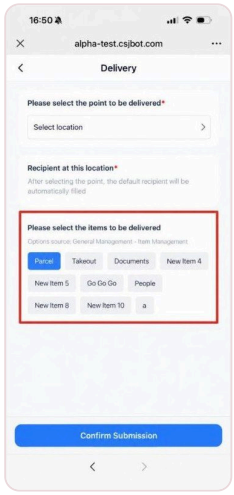
Retrieve before delivery shows an error

2 Choose destination, recipient and item type

Tap **Select location** and pick a point → the default recipient fills in (edit up to 3) → choose the item type (Parcel / Takeaway / Documents ...).



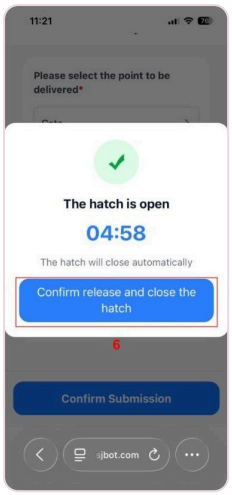
Delivery steps 1–5



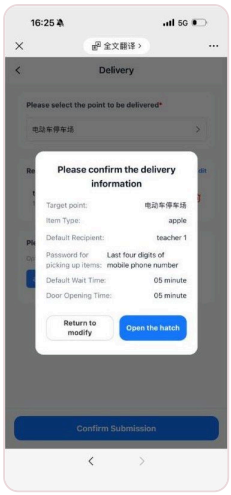
Choose item type

3 Open, load and send

Tap **Open the hatch**, load the item securely so it does not block the lid, then do step 6 (close). The robot closes the lid and drives off.



Step 6: close and dispatch



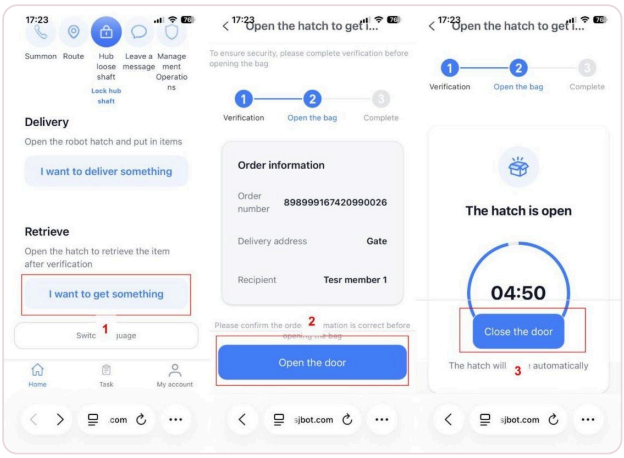
Delivery confirmation

5.5 Pick up an item (Retrieve)

1

Open Retrieve

When the robot arrives, tap **I want to get something** (Retrieve) on Home.

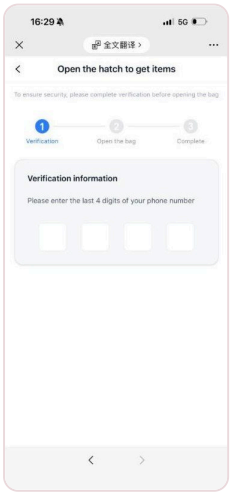


Retrieve steps 1–3

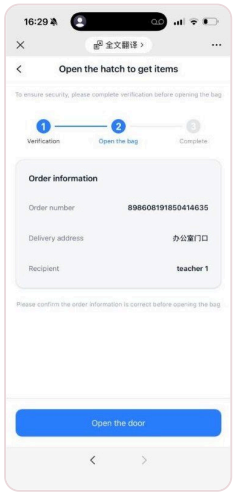
2

Verify

Enter the last 4 digits of the recipient's phone or the pickup password, tap **Verify** → check the order → **Open the door**.



Verification

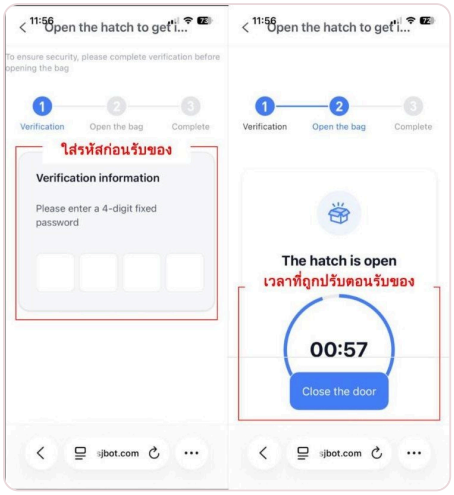


Order information

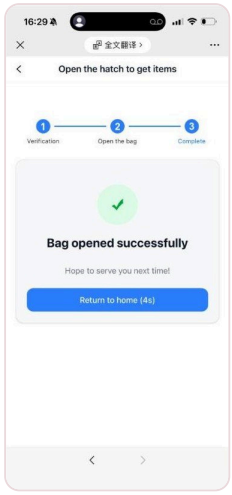
3 Take the item and close

Take the item before the countdown ends, then tap **Close the door** to finish. The app returns to Home.

> ⚠ **Caution** Do not force or strike the hatch; contact the admin if it does not open



Verification and countdown

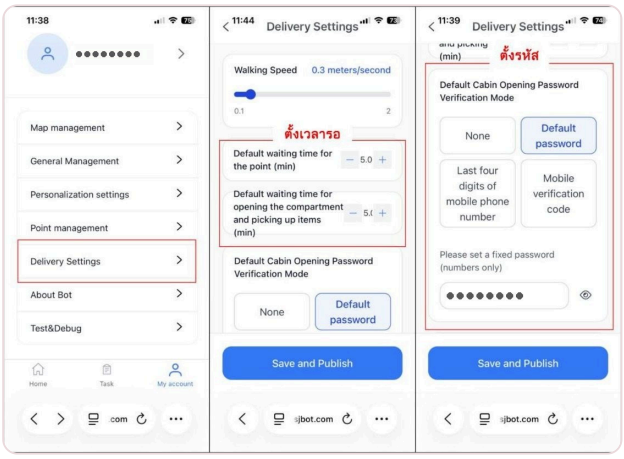


Pickup successful

5.6 Extra delivery settings

1 Wait time and pickup password

In Delivery settings set the wait time at the point — **after timeout the robot cancels the task and will not open; start again** · choose the opening method: None / Default password / last 4 phone digits / SMS code.

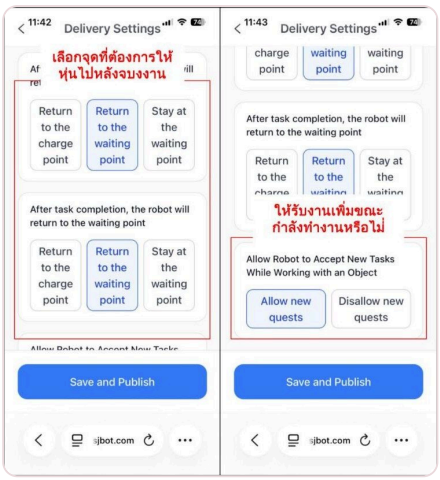


Wait time and password

2

Return point and new tasks

Choose whether the robot returns to the **Charging point** or **Waiting point** after a task, and whether it accepts new tasks while items remain inside, then **Save and Publish**.



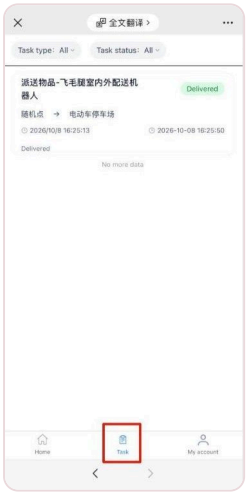
Return point and new tasks

5.7 Track tasks

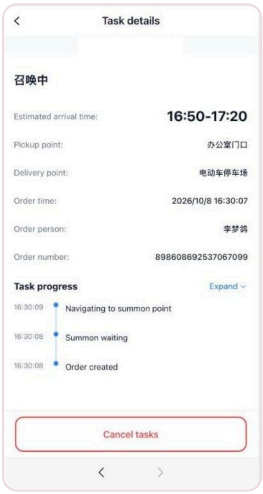
1

Task list

The **Task** tab lists all tasks with type/status filters. Tap a task for details, map, progress and cancel.



Task tab



Task details

5.8 Admin controls (Management Operations)

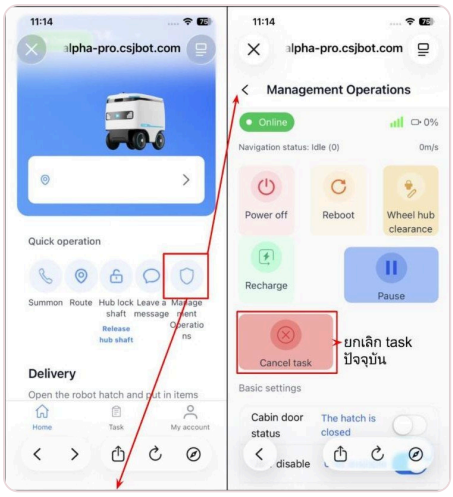
1

Open Management Operations

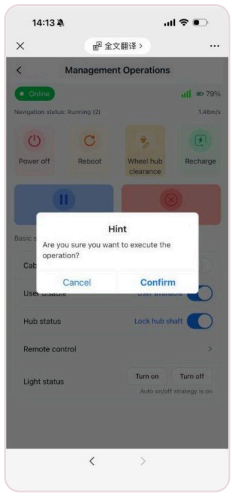
On Home tap **Management Operations**. Every command asks for confirmation.

Button	What it does
Power off	Remote shutdown (only when idle and safe)
Reboot	Restart when unresponsive or abnormal
Wheel hub clearance	Clear wheel faults (keep people away)
Recharge	Start / cancel return to charge

Button	What it does
Pause	Pause the current task
Cancel task	Cancel the task (handle any items first)



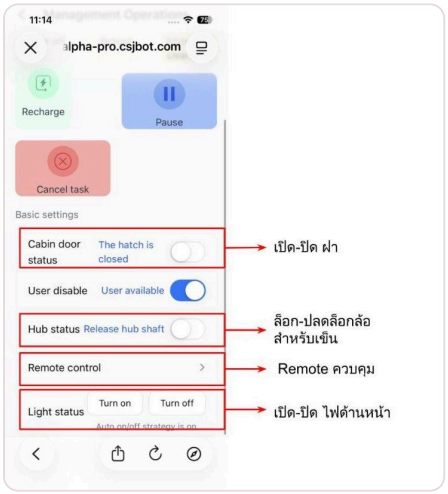
Menu entry and cancel task



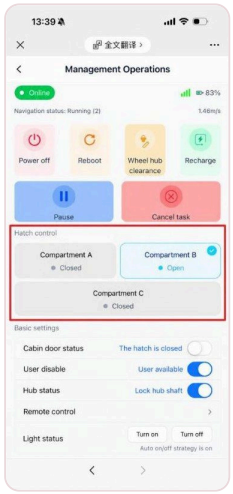
Confirmation dialog

2 Basic settings

Open/close the hatch (Cabin door / Hatch control A–B–C), enable/disable users, lock/release the hub, front light on/off.



Door, hub, remote and front light

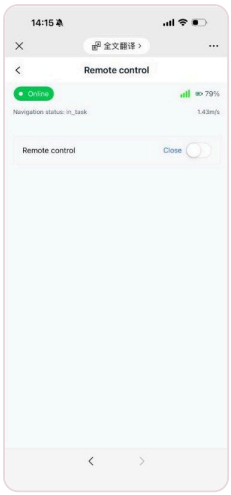


Hatch control

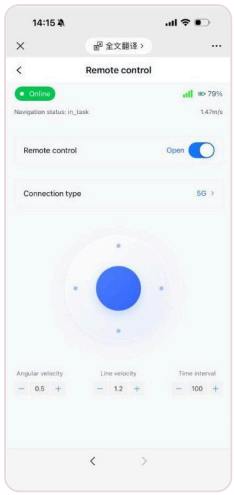
3 Remote control

Tap **Remote control** → switch it on → use the joystick to drive; adjust angular/linear velocity with – / +. **Switch it off** when done.

> ⚠ **Caution** Collision risk — only with the robot in sight, nobody nearby, a clear path and low speed



Remote control off

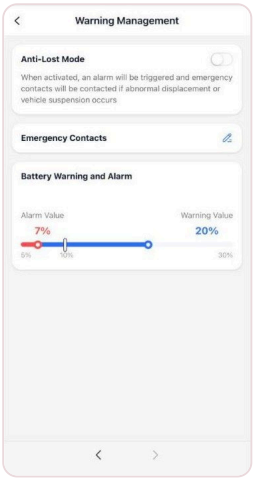


Remote control on

4

Battery alerts and emergency contacts

My account → Warning management: anti-lost mode, up to 3 emergency contacts, warning and critical battery levels (critical must be lower).



Warning management

06

Testing & hand-over

A step-by-step test before going live, indoors and outdoors, plus checklist, troubleshooting and maintenance

▶ VIDEO TUTORIAL

06 · Outdoor field test

https://drive.google.com/file/d/19uW-sWlgD4ksZYHdVEgyuLAABZoas5ji/view

Scan to watch

6.1 Test procedure

1 **Pre-check**

Robot online (green), battery above 50%, status in_idle, hub locked, hatches closed, speed set low.

2 **Localization**

Restart at the origin and confirm the position and heading on the map match reality.

3 **Drive to every point**

Use **Point management** → **Go** for each point; check it parks accurately, avoids kerbs and respects virtual walls.

4 **Obstacle avoidance**

Have a person step in front or place a box in the path at a safe distance; the robot must slow/stop and route around.

5 **End-to-end task**

Summon → Delivery (load) → Retrieve (verify and unload) → robot returns to the waiting point · time it and check recipient notifications.

6 **Exception cases**

Pause / cancel mid-route, let the pickup wait expire (the robot must cancel itself), try a wrong code, short remote-control drive.

7 **Outdoors (field test)**

Check surfaces, ramps, kerbs, 4G/5G dead spots and weather. Run the real route at least 2–3 times with someone following.

6.2 Hand-over checklist

- [ ] SIM, battery and side cover fitted; robot online
- [ ] Robot WiFi and Deployment login work
- [ ] Whole area mapped; returned to origin within 20 cm, same heading
- [ ] Map edited (noise, virtual walls, routes if needed) and saved
- [ ] Map synced, set as default and robot restarted at the origin
- [ ] All points added (incl. waiting point) and Synchronization points done
- [ ] Go to every point passed
- [ ] Summon / Delivery / Retrieve end-to-end passed
- [ ] Wait times, pickup code, return point, recipients and emergency contacts set
- [ ] Users trained on charging, hub lock/release and emergency stop

6.3 Troubleshooting

Problem	Fix
Mapping initialization hangs	Restart at the origin, reconnect WiFi, wait ~5 min and start again
Wheels do not move in remote mode	Make sure the hub is locked before remote control
WiFi drops while mapping	Move closer and continue once the signal returns; remap if lost for long
Edits not shown after saving	Confirm Save and finish editing was clicked, wait, then exit and re-enter
New map not in use	Check it is the default map and the robot was restarted at that map's origin
Robot cannot reach a point	Sync points again, check the right map is selected and the point is in open space away from virtual walls
Retrieve shows an error	There must be a delivery that has arrived before you can retrieve
Hatch will not open / task cancelled	The wait time probably expired; start again or let an admin open the hatch from Management Operations
Robot offline (grey)	Check battery, SIM/4G signal and restart the robot
Fault (red)	Stop tasks, try Wheel hub clearance or Reboot; if it persists send TESR photos/video

6.4 Care and transport

- Check online state, battery and pending tasks every day; charge when low

- Keep the charging area dry and clear; do not move the charger without permission
- Clean the screen, QR area, compartments and wheels regularly
- Short moves: push on flat floor ≤30 m/min with power off or hub released · vehicles, stairs or steps over 5 mm need the vendor's handling tool and two people
- Long-distance: use the vendor box and padding, strap the robot down (cargo height over 1.5 m) with 2 cm foam

Contact TESR

TESR Co., Ltd. (Thai Embedded System and Robotics) · tesrshop.com · tesracademy.com — report issues with the robot SN, photos/video and the steps that led to the problem.

Compiled from: TESR Thai setup notes, the vendor's Speedybot H5 (New Version) Product Manual V1.1.0 and the Indoor/Outdoor Speedbot Mapping Deployment V1.0 Operation Guide